

Georgia State University
CSC 4120 Introduction to Robotics

Whiteboard Eraser Robot

Code & Hardware

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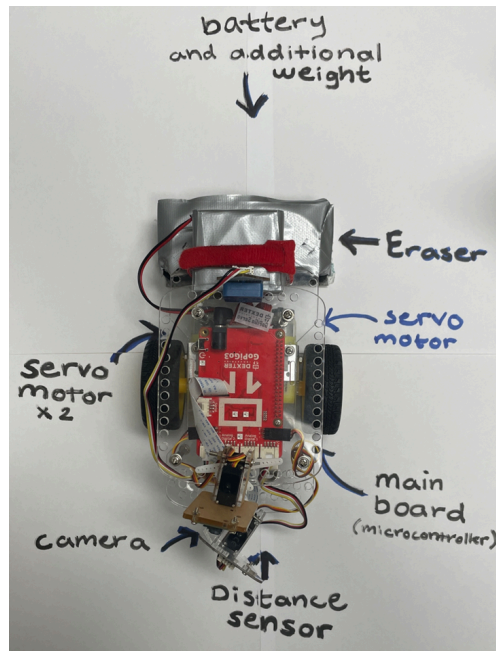
I. Problem Statement

In educational settings, the manual task of erasing whiteboards can interrupt the flow of lectures, reduce class time, and leave residual marks that obscure new material. This recurring issue disrupts the teaching process and affects student engagement. To address this problem, an automated solution is needed to maintain a clear whiteboard surface without manual intervention, allowing instructors to focus on delivering uninterrupted lessons.

II. Approach

The whiteboard-erasing robot aims to clean whiteboards efficiently by rotating a specialized erasing mechanism to remove marker traces. Instead of relying on computer vision, the robot will cover the entire whiteboard by following a systematic path, ensuring thorough cleaning. It will detect the edges of the board to determine its dimensions and adjust its movement to accommodate various board sizes and shapes. The rotating eraser mechanism will ensure even cleaning across the surface, leaving no streaks or smudges. Designed for minimal human intervention, the robot will provide a convenient, hands-free solution for maintaining clear whiteboards in different classroom and office environments.

III. Design Detail



This image depicts the Whiteboard Erasing Robot from a top-down view, and each labeled component is part of the robot's key functionality. (Battery, Eraser, Servo Motors, Main Board, Camera, Distance Sensor)

- Servo Motors: One servo motor is likely responsible for moving the robot forward or backward, while the other helps with directional adjustments or turning.
- Main Board: The central processing unit or microcontroller of the robot, where all signals from the sensors are processed. It likely controls the motors, camera, and sensors. The main board coordinates the robot's movement, the erasing mechanism, and the computer vision processing.
- Camera: This is the key component of the robot's computer vision system. The camera detects dirty or marked areas on the whiteboard, distinguishing them from clean areas. It sends this information to the main board for processing, helping the robot know where to erase.
- Distance Sensor: Positioned near the front of the robot, the distance sensor helps the robot detect boundaries, such as the edges of the whiteboard. It ensures that the robot doesn't move off the whiteboard or into any obstructed areas. It's crucial for safe and effective navigation, ensuring the robot stays within the workspace.
- Battery: Powers the motors and control board

IV. System Requirements

- Edge Detection: The sandbox-like edge acts as a physical and visual boundary that prevents the robot from navigating outside of the whiteboard area. The edge detection system, utilizing the robot's camera and sensors, ensures the robot stays within the board's perimeter and avoids unnecessary movements. To ensure full coverage of the whiteboard, we've divided the board into four quadrants. This allows the robot to methodically clean one section at a time, ensuring no area is missed. The robot will execute movement patterns within each quadrant, optimizing coverage while minimizing overlap. This quadrant-based system ensures that the robot efficiently navigates each part of the board while accounting for the edge boundaries.

- Adaptive Cleaning: The robot's erasing by mechanism will adapt based on the condition of the whiteboard's surface. Depending on the type and level of residue present, the system will adjust the cleaning intensity and erasing pattern.
- Coverage: The Whiteboard Erasing Robot must ensure that all areas of the whiteboard are fully cleaned during operation. This includes using an efficient path-planning algorithm to cover the board without missing sections or wasting time on already cleaned areas.

V. Project Requirements

- Software
 - Python 3.x
 - OpenCV
 - NumPy
 - RPLidar Python SDK
 - Imutils
- Hardware
 - Servo motors (x2)
 - LiDAR sensor
 - Camera module
 - Battery pack
 - Microcontroller (e.g., Raspberry Pi)

VI. How to Use the Program

1. Start the Robot
 - a. Turn on the gopigo robot. Within 10–30 seconds, a green light will illuminate. After this, connect your computer to the gopigo Wi-Fi network.
2. Verify Connection
 - a. Once the gopigo successfully connects to your computer, a blue light will turn on.
3. Access Control Interface
 - a. Open your browser and go to mygopigo.com to control the robot.

4. Activate Erasing Functionality
 - a. Click on the side panel of the site and connect the distance sensor to IC2-1.
 - b. Open the terminal and run the following command: `python3 gopigo_drive.py`
 - c. As the code executes, I observe the whiteboard eraser in action, effectively detecting and removing the mark on the whiteboard with precision.

Github: <https://github.com/hamchinny/batman>